

CLEVELAND BURKE LAKEFRONT, OH, USA

WMO: 725245

Lat: 41.518N

Lon: 81.684W

Elev: 584

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.4	(c) 12.0	(d) -2.3	(e) 5.0	(f) 8.8	(g) 2.3	(h) 6.3	(i) 13.8	(j) 33.5	(k) 27.9	(l) 31.1	(m) 24.1	(n) 13.3	(o) 220	(p) 0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 88.8	(d) 73.4	(e) 85.9	(f) 72.6	(g) 83.5	(h) 72.0	(i) 76.3	(j) 84.2	(k) 75.0	(l) 82.5	(m) 73.8	(n) 81.2	(o) 11.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	126.9	81.0	72.4	122.8	80.3	71.5	118.9	79.7	40.0	84.5	38.8	82.6	37.6	81.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.4	(b) 25.3	(c) 22.7	(e) 3.3	(f) 94.1	(g) 6.8	(h) 2.4	(i) -1.6	(j) 95.8	(k) -5.5	(l) 97.2	(m) -9.3	(n) 98.6	(o) -14.3	(p) 100.3		
(a) 28.4	(b) 25.3	(c) 22.7	(e) 2.2	(f) 78.7	(g) 6.6	(h) 1.8	(i) -2.6	(j) 80.1	(k) -6.5	(l) 81.1	(m) -10.2	(n) 82.2	(o) -15.0	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	29.6	31.4	38.3	50.0	61.4	70.6	74.7	73.7	67.7	56.3	45.9	35.3
	DBStd	18.20	11.04	10.71	10.61	9.17	8.60	6.82	4.89	4.58	6.68	8.09	8.56	9.64
	HDD50	2349	637	527	394	111	6	0	0	0	0	28	180	466
	HDD65	5361	1097	940	829	461	179	25	1	1	46	290	573	919
	CDD50	3451	5	6	31	110	358	619	767	733	531	223	57	11
	CDD65	993	0	0	2	9	66	194	303	270	127	21	1	0
	CDH74	6466	0	0	8	91	434	1289	2204	1628	702	108	2	0
	CDH80	1561	0	0	0	12	102	375	550	341	169	12	0	0
Wind	WSAvg	10.9	13.6	12.4	11.7	11.3	9.5	8.7	8.3	8.6	9.4	11.5	12.6	13.3
Precipitation	PrecAvg	40.3	3.0	2.3	2.8	3.5	3.7	3.8	4.2	3.5	3.9	3.5	2.9	3.1
	PrecMax	52.5	6.1	5.5	6.0	5.8	7.3	8.0	9.3	7.8	9.0	6.8	5.0	7.1
	PrecMin	28.2	1.5	0.9	0.6	1.7	1.4	0.8	1.3	1.3	1.2	1.5	1.2	1.6
	PrecStd	5.7	1.1	1.0	1.1	1.1	1.7	1.7	1.9	1.4	1.7	1.2	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	67.8	74.9	81.7	86.9	91.5	91.7	91.2	89.8	81.4	71.9	65.1
		MCWB	56.4	56.7	59.7	63.3	70.4	73.0	75.3	73.6	71.9	67.7	60.2	57.2
	2%	DB	56.8	58.5	67.8	76.5	82.8	87.9	88.2	86.7	84.4	77.0	66.2	59.1
		MCWB	51.8	51.4	56.4	61.6	69.1	72.6	74.7	73.6	71.4	65.3	57.2	52.9
	5%	DB	51.2	52.0	62.4	71.4	79.2	84.1	85.0	83.5	81.2	72.8	63.1	54.0
		MCWB	45.6	45.2	53.2	58.2	66.4	71.4	73.5	72.8	70.0	62.8	54.8	49.1
	10%	DB	45.0	45.6	54.8	65.5	74.9	81.2	82.6	81.3	78.0	68.7	59.2	48.3
		MCWB	40.0	39.9	48.2	55.8	64.6	70.2	72.6	71.6	68.7	60.4	51.8	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	58.3	61.4	66.6	73.2	76.3	79.1	77.4	75.5	70.6	63.0	59.1
		MCDB	62.0	66.1	72.4	77.3	83.5	85.5	87.2	85.7	83.9	79.4	68.1	63.6
	2%	WB	52.4	52.1	58.0	63.3	70.5	74.5	77.0	75.6	73.6	67.7	59.1	54.1
		MCDB	56.4	57.4	66.6	73.7	80.4	83.6	85.1	83.0	80.9	75.0	65.1	58.6
	5%	WB	46.2	45.5	53.3	59.7	68.2	73.0	75.4	74.4	71.7	64.4	56.0	48.9
		MCDB	50.1	50.5	60.4	69.5	77.0	82.1	82.4	81.3	78.6	70.3	61.3	53.0
	10%	WB	40.0	40.1	48.4	56.1	66.0	71.2	74.1	73.0	69.8	61.5	52.7	44.2
		MCDB	44.9	45.6	55.4	64.4	74.0	79.5	81.1	79.7	76.6	67.8	58.2	48.0
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.3	13.0	14.6	14.1	12.7	11.8	11.8	12.8	12.5	11.7	10.1
		MCDBR	19.0	21.7	24.3	24.5	19.7	17.3	15.1	15.0	16.6	17.8	17.6	16.4
		MCWBR	16.1	17.0	16.5	14.7	11.8	9.3	8.9	8.6	9.6	10.8	12.8	13.4
	5% WB	MCDBR	18.7	20.9	22.8	23.2	17.7	14.8	12.3	12.8	13.9	15.0	15.9	16.1
		MCWBR	16.9	17.4	16.4	15.5	11.1	8.7	8.4	8.4	9.0	10.3	12.9	14.6
Clear-Sky Solar Irradiance	taub	0.308	0.337	0.361	0.400	0.436	0.454	0.454	0.438	0.390	0.371	0.329	0.311	
	taud	2.398	2.273	2.288	2.269	2.202	2.196	2.221	2.253	2.362	2.392	2.483	2.473	
	Ebn at Noon	266	273	280	276	268	262	261	261	267	259	257	257	
	Edh at Noon	26	34	37	41	45	45	44	41	35	30	24	22	
All-Sky Solar Radiation	RadAvg	431	707	1035	1416	1702	1890	1896	1675	1334	848	533	348	
	RadStd	39	89	103	124	147	141	121	78	121	74	85	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	N/A

Nomenclature: See separate page